

Copper Reflectance Model—BBRsim Brief

Golwala / Mirabolfathi group

Contents

1	Conductivity	1
2	Drude Scattering Time	2
3	Drude AC Conductivity and Taylor Approximations	2
4	Complex Wave Vector (Griffiths §9.4)	3
5	Reflectance and Absorptance	3
6	Hagen-Rubens Limit and Breakdown Frequency	4
7	Regime Summary	5
8	Copper Material Constants	5
9	Summary of Inputs	5
10	Figures	5

1 Conductivity

The room-temperature conductivity of copper is universal across all grades:

$$\sigma_{\text{RT}} = 5.96 \times 10^7 \text{ S/m}$$

At cryogenic temperature T with residual resistance ratio RRR, Matthiessen's rule gives the total scattering rate as the sum of impurity and phonon contributions:

$$\frac{1}{\sigma(T, \text{RRR})} = \frac{1}{\sigma_{\text{impurity}}} + \frac{1}{\sigma_{\text{phonon}}(T)}, \quad \sigma_{\text{impurity}} = \text{RRR} \times \sigma_{\text{RT}}, \quad \sigma_{\text{phonon}}(T) \approx \sigma_{\text{RT}} \frac{273 \text{ K}}{T}.$$

At $T = 4 \text{ K}$ phonon scattering is frozen out and $\sigma_{\text{DC}}(4 \text{ K}) \simeq \text{RRR} \times \sigma_{\text{RT}}$. At $T = 273 \text{ K}$ the result is σ_{RT} regardless of RRR, consistent with the RRR definition.

The **plasma frequency** of copper (free-electron model):

$$\omega_p = \sqrt{\frac{n_e e^2}{\epsilon_0 m_e}} \approx 1.64 \times 10^{16} \text{ rad/s} \implies f_p \approx 2.6 \text{ PHz (UV)},$$

where the material constants are listed in Section 8. We will see that ω_p sets the upper limit of metallic reflection.

2 Drude Scattering Time

The electron scattering time τ connects the DC conductivity to the free-electron Drude picture:

$$\tau = \frac{\sigma_{\text{DC}} m_e}{n_e e^2} = \frac{\sigma_{\text{DC}}}{\varepsilon_0 \omega_p^2}.$$

A useful identity: $\sigma_{\text{DC}} = \varepsilon_0 \omega_p^2 \tau$.

Symbol	Value	Meaning
n_e	$8.49 \times 10^{28} \text{ m}^{-3}$	free electron density
m_e	$9.109 \times 10^{-31} \text{ kg}$	electron mass
e	$1.602 \times 10^{-19} \text{ C}$	electron charge
ω_p	$1.64 \times 10^{16} \text{ rad/s}$	plasma angular frequency

For OFHC Cu (RRR = 100) at 4 K: $\sigma_{\text{DC}} = 5.96 \times 10^9 \text{ S/m}$ and $\tau \approx 2.5 \text{ ps}$.

3 Drude AC Conductivity and Taylor Approximations

The frequency-dependent (AC) conductivity from the Drude model:

$$\sigma(\omega) = \frac{\sigma_{\text{DC}}}{1 - i\omega\tau} = \frac{\sigma_{\text{DC}}(1 + i\omega\tau)}{1 + \omega^2\tau^2}.$$

The real and imaginary parts are:

$$\text{Re}[\sigma(\omega)] = \frac{\sigma_{\text{DC}}}{1 + (\omega\tau)^2}, \quad \text{Im}[\sigma(\omega)] = \frac{\sigma_{\text{DC}} \omega\tau}{1 + (\omega\tau)^2}.$$

$\text{Im}[\sigma]$ peaks at $\omega\tau = 1$ with value $\sigma_{\text{DC}}/2$.

Low-frequency limit ($\omega\tau \ll 1$): Taylor series

Expanding $\sigma(\omega) = \sigma_{\text{DC}} \sum_{n=0}^{\infty} (i\omega\tau)^n$:

$$\begin{aligned} \text{Re}[\sigma(\omega)] &= \sigma_{\text{DC}} [1 - (\omega\tau)^2 + (\omega\tau)^4 - \dots], \\ \text{Im}[\sigma(\omega)] &= \sigma_{\text{DC}} [\omega\tau - (\omega\tau)^3 + \dots]. \end{aligned}$$

At leading order $\sigma(\omega) \rightarrow \sigma_{\text{DC}}$: the DC conductivity fully determines the optical response, giving the Hagen-Rubens limit.

High-frequency limit ($\omega\tau \gg 1$): Laurent series

Writing $1/(1 - i\omega\tau) = (i/\omega\tau)/(1 + i/\omega\tau)$ and expanding:

$$\begin{aligned} \text{Re}[\sigma(\omega)] &= \frac{\sigma_{\text{DC}}}{(\omega\tau)^2} \left[1 - \frac{1}{(\omega\tau)^2} + \dots \right] \rightarrow \frac{\sigma_{\text{DC}}}{(\omega\tau)^2}, \\ \text{Im}[\sigma(\omega)] &= \frac{\sigma_{\text{DC}}}{\omega\tau} \left[1 - \frac{1}{(\omega\tau)^2} + \dots \right] \rightarrow \frac{\sigma_{\text{DC}}}{\omega\tau} = \frac{n_e e^2}{m_e \omega} = \frac{\varepsilon_0 \omega_p^2}{\omega}. \end{aligned}$$

In this regime the conductivity is almost purely imaginary (inductive). The real part $\text{Re}[\sigma] \propto (\omega\tau)^{-2}$ is the residual dissipation. The imaginary part at leading order $\varepsilon_0\omega_p^2/\omega$ is frequency-dependent and independent of τ — it encodes the inertia of the electron gas, not its damping.

4 Complex Wave Vector (Griffiths §9.4)

For a plane wave in a medium with real permittivity ε_0 and conductivity $\sigma_r = \text{Re}[\sigma(\omega)]$, Griffiths eqs. 9.126 give the complex wave vector $\tilde{k} = k + i\kappa$:

$$k = \frac{\omega}{c} \sqrt{\frac{1}{2} \left(\sqrt{1 + \left(\frac{\sigma_r}{\varepsilon_0\omega} \right)^2} + 1 \right)}, \quad \kappa = \frac{\omega}{c} \sqrt{\frac{1}{2} \left(\sqrt{1 + \left(\frac{\sigma_r}{\varepsilon_0\omega} \right)^2} - 1 \right)}.$$

The complex refractive index $\tilde{n} = n + ik_{\text{opt}} = c\tilde{k}/\omega$, so $n = ck/\omega$ and $k_{\text{opt}} = c\kappa/\omega$.

Regime approximations

Throughout the BBRsim range (50 GHz–20 THz) for OFHC Cu at 4 K, the ratio $\sigma_r/(\varepsilon_0\omega) \gg 1$ (it reaches $\sim 10^5$ at 1 THz). In this **good-conductor limit**:

$$k \approx \kappa \approx \frac{\omega}{c} \sqrt{\frac{\sigma_r}{2\varepsilon_0\omega}} = \sqrt{\frac{\mu_0\sigma_r\omega}{2}}.$$

The skin depth $\delta = 1/\kappa$.

Low-f ($\omega\tau \ll 1$, *H-R regime*): $\sigma_r \approx \sigma_{\text{DC}}$, so

$$k \approx \kappa \approx \sqrt{\frac{\mu_0\sigma_{\text{DC}}\omega}{2}}, \quad n \approx k_{\text{opt}} \approx \sqrt{\frac{\sigma_{\text{DC}}}{2\varepsilon_0\omega}} \gg 1.$$

High-f ($\omega\tau \gg 1$, *inductive regime*): $\sigma_r \approx \sigma_{\text{DC}}/(\omega\tau)^2$, so

$$k \approx \kappa \approx \frac{1}{\omega\tau} \sqrt{\frac{\mu_0\sigma_{\text{DC}}\omega}{2}}, \quad n \approx k_{\text{opt}} \approx \frac{n_{\text{HR}}}{\omega\tau},$$

where $n_{\text{HR}} = \sqrt{\sigma_{\text{DC}}/(2\varepsilon_0\omega)}$ is the H-R refractive index. The skin depth grows as $\delta \propto \omega^{-1/2} \cdot \omega\tau$, i.e. the metal becomes more transparent above f_{break} .

5 Reflectance and Absorptance

Normal-incidence Fresnel reflectance from vacuum into the conductor:

$$R = \left| \frac{\tilde{n} - 1}{\tilde{n} + 1} \right|^2 = \frac{(n - 1)^2 + k_{\text{opt}}^2}{(n + 1)^2 + k_{\text{opt}}^2}, \quad D = 1 - R.$$

For $n \approx k_{\text{opt}} \gg 1$ (good-conductor limit): $D \approx 2/n = 2\sqrt{2\varepsilon_0\omega/\sigma_r}$.

Universal Drude formula

Inserting $\sigma_r = \sigma_{\text{DC}}/(1 + (\omega\tau)^2)$ into $D \approx 2\sqrt{2\varepsilon_0\omega/\sigma_r}$:

$$D(\omega) \approx 2\sqrt{\frac{2\varepsilon_0\omega}{\sigma_{\text{DC}}}} \cdot \sqrt{1 + (\omega\tau)^2} = D_{\text{HR}}(\sigma_{\text{DC}}) \cdot \sqrt{1 + (\omega\tau)^2}.$$

This single expression is valid throughout the BBRsim range. It connects the H-R limit continuously to the inductive regime through the single dimensionless parameter $\omega\tau$.

Taylor and Laurent expansions of D

Low-f ($\omega\tau \ll 1$):

$$D \approx D_{\text{HR}} \left[1 + \frac{1}{2}(\omega\tau)^2 - \frac{1}{8}(\omega\tau)^4 + \dots \right], \quad D \propto \omega^{1/2}.$$

High-f ($\omega\tau \gg 1$):

$$D \approx D_{\text{HR}} \cdot \omega\tau \left[1 + \frac{1}{2(\omega\tau)^2} + \dots \right], \quad D \propto \omega^{3/2}.$$

Above f_{break} the absorptance rises *faster* than the H-R prediction ($\omega^{3/2}$ vs. $\omega^{1/2}$) because the reduced σ_r at higher frequency allows deeper penetration.

6 Hagen-Rubens Limit and Breakdown Frequency

In the limit $\omega\tau \ll 1$ (DC approximation), Eq. (5) reduces to:

$$D = 1 - R \approx 2\sqrt{\frac{2\varepsilon_0\omega}{\sigma_{\text{DC}}}}.$$

This limit breaks down above the scattering frequency:

$$f_{\text{break}} = \frac{1}{2\pi\tau} = \frac{n_e e^2}{2\pi m_e \sigma_{\text{DC}}}.$$

Grade	RRR	σ_{DC} [S/m]	τ [ps]	f_{break} [GHz]
Commercial Cu	10	5.96×10^8	0.25	640
OFHC Cu	100	5.96×10^9	2.5	64
H ₂ -annealed	57	3.4×10^9	1.4	115
Ultra-pure crystal	500	3.0×10^{10}	12.5	13

BBRsim spans 50 GHz–20 THz. Hagen-Rubens is inadequate for essentially the entire range at cryogenic temperatures: even for commercial Cu (RRR = 10, $f_{\text{break}} = 640$ GHz) $\sim 97\%$ of the range lies above the limit. **The full Drude model is always required.**

7 Regime Summary

Regime	Condition	$D \approx$	log-log slope
Hagen-Rubens (DC)	$\omega\tau \ll 1$	$2\sqrt{2\varepsilon_0\omega/\sigma_{\text{DC}}}$	$\frac{1}{2}$
Drude (full)	$\omega\tau \sim 1$	$D_{\text{HR}}\sqrt{1 + (\omega\tau)^2}$	$\frac{1}{2} \rightarrow \frac{3}{2}$
Inductive	$\omega\tau \gg 1, \omega \ll \omega_p$	$2\omega\tau\sqrt{2\varepsilon_0\omega/\sigma_{\text{DC}}}$	$\frac{3}{2}$
Plasma threshold	$\omega \rightarrow \omega_p$	$R \rightarrow 0$ (transparent)	—

For copper, $f_p \approx 2.6$ PHz (ultraviolet), so the plasma threshold is far above the BBRsim range. The relevant transition in BBRsim is H-R $\rightarrow$ inductive, driven by f_{break} .

8 Copper Material Constants

Constant	Symbol	Value	Units
Room-temperature conductivity	σ_{RT}	5.96×10^7	S/m
Free electron density	n_e	8.49×10^{28}	m ⁻³
Electron mass	m_e	9.109×10^{-31}	kg
Electron charge	e	1.602×10^{-19}	C
Plasma frequency	ω_p	1.64×10^{16}	rad/s
Debye temperature	Θ_D	343	K

σ_{RT} is universal for all copper grades.

9 Summary of Inputs

All other quantities ($\sigma_{\text{DC}}, \tau, \tilde{n}, R$) are derived from just three inputs:

Parameter	Meaning
RRR	Residual resistance ratio; characterises Cu purity
T [K]	Temperature of the copper surface
$\omega = 2\pi f$	Photon angular frequency

10 Figures

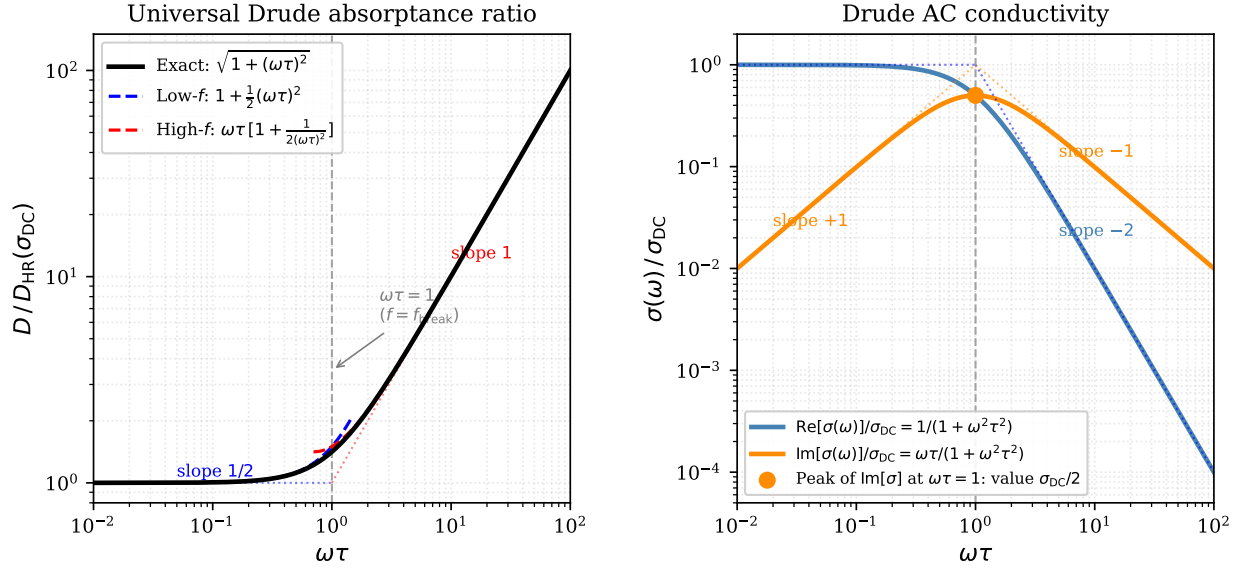


Figure 1: *Left*: Universal Drude absorptance ratio D/D_{HR} vs. dimensionless frequency $\omega\tau$. At $\omega\tau \ll 1$ (Hagen-Rubens regime), $D/D_{\text{HR}} \rightarrow 1$. Above $\omega\tau = 1$ ($f = f_{\text{break}}$), D/D_{HR} grows as $\omega\tau$, changing the log-log slope from $\frac{1}{2}$ to $\frac{3}{2}$. The exact closed form $\sqrt{1 + (\omega\tau)^2}$ (solid line) and the two-term Taylor/Laurent approximations (dashed) are shown. *Right*: Real and imaginary parts of $\sigma(\omega)/\sigma_{\text{DC}}$ vs. $\omega\tau$. $\text{Re}[\sigma]$ is a Lorentzian that rolls off as $(\omega\tau)^{-2}$; $\text{Im}[\sigma]$ peaks at $\omega\tau = 1$ with value $\sigma_{\text{DC}}/2$, then falls as $(\omega\tau)^{-1}$. Dotted lines indicate the asymptotic power laws.

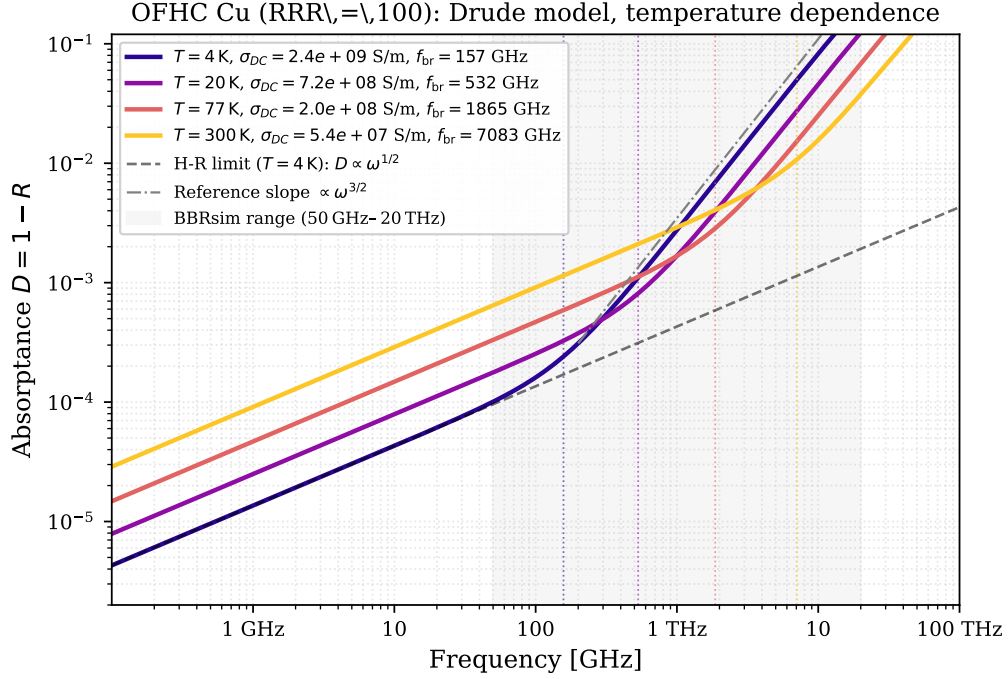


Figure 2: Absorptance $D = 1 - R$ vs. frequency for OFHC Cu (RRR = 100) at $T = 4, 20, 77, 300$ K (Drude model). Dotted vertical lines mark $f_{\text{break}} = 1/(2\pi\tau)$ for each temperature; the slope visibly steepens from $\frac{1}{2}$ (H-R) to $\frac{3}{2}$ (inductive) above each line. The dashed black line is the H-R approximation for $T = 4$ K, illustrating where it diverges from the full model. The grey dash-dot line shows the $\omega^{3/2}$ reference slope. The grey band is the BBRsim simulation range (50 GHz–20 THz).